

BOTANY XII

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF BIOLOGY.

COMPILED BY Dr. Abdul Raheem

0311-3136863

QUESTIONS		YEARS
Chapter: 22	CHROMOSOME AND DNA	
Section B		
i) Give the name of enzymes involved in replication of DNA, also give function of any one of them. ii) Differentiate between protein synthesis in Prokaryotes and Eukaryotes. viii) What is heteroploidy? Write its types.		2025
xi) Write a note on any one of the following: * Sickle Cell Anemia * Phenylketonuria		2024
(i) What do you know about sickle cell Anemia OR Alkaptonuria? (iii) State Hershey and Chase experiment that confirmed DNA as a hereditary material. (vi) What are the effects of Chemical Mutagens on DNA?		2023
2. ii) Name various Nitrogen bases of DNA. 2. iii) Differentiate any one of the following: (a) DNA and RNA (b) Autosome and Sex Chromosome 3. ii) Draw a chart to show the chemical composition of chromosome.		2022
2. iii) Mention number of autosomes and sex chromosomes in a human cell 2. vii) Name various types of RNA involved in protein biosynthesis. 3. iv) Mention different types of chromosomes according to the position of centrometre.		2021

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3.vi) Define mutation. How does ionization radiation damage the DNA?	
2. (vi) Draw Karyotype of Drosophila. 2. (xiii) Name various nitrogenous bases found in DNA nucleotides. 3. (iii) Classify chromosome on the basis of position of centromere.	2019
3.iii) What is an autosome? How many Autosomes are there in a human cell?	2018
2.iii) Name the compounds of DNA. 2.iv) Name the scientist known for the following discoveries. (a) ladder type of organization of DNA (b) One gene-one enzyme hypothesis 3. viii) Write note on genetic code. 3. vii) Define merism and mention its types OR Differentiate between DNA and R.N.A.	2017
2. i) Differentiate any One of the following: a) DNA and RNA b) Autosome and Sex-chromosomes 2. viii) Write a note Phenylketonuria Or Alkaptonuria. 2. xiii) Write a note on genetic code.	2016
iii) Explain Chromosomal Aberration. vi) Describe the types of RNA.	2015
2. ii) Differentiate between DNA and RNA. 2. xi) State four point of Chromosomal theory of heredity. 2. xii) Write a note on the significance of Secondary growth.	2014
Section C	
3. Give the evidence of DNA as hereditary material.	2025
4. Explain the term DNA replication. Describe the process of DNA replication with diagram.	2024
4. —OR— Describe the Law of Independent Assortment with the help of genetic cross.	
4. Describe the ultra-structure of chromosome with their chemical composition. Also discuss its types.	2023
6. a) Describe Watson and Crick model of DNA.	2022
7. Give a brief description of structure of DNA.	2021
4. Describe Watson and Crick model of DNA. How does the model explain semi-conservative replication of DNA?	2019
4) Prove with the help of experiment that DNA is a genetic material.	2017

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4 OR] Explain Watson & crick model of DNA and write its replication.	
4. Define Chromosome. Describe with diagram the chemical composition of Chromosome and their ultra-structure.	2016
3. Describe Watson and Cricks Model of DNA. How does DNA replicates semi-conservatively.	2015
Chapter: 23	INHERITANCE
Section B	
iii) Why human males are heterogametic? x) Why Y-linked traits are called holandric Traits? ix) Explain any one of the following: *co-dominance *incomplete dominance	2025
x) Who O-negative blood group is called the universal donor? xii) Write a note on any one of the following: * Incomplete Dominance * Co-Dominance * Test Cross xiii) Why haemophilia is common in male?	2024
2. iv) What is a gene?	2022
3. i) What is Test Cross? Why is it performed?	
3.i) State the law of segregation?	2021
3.iii) Make a cross between a color blind male and normal female and mention result.	
2. (xiv) Which cross helps to determine the homozygosity of a dominant parent?	2019
2.x) Define test cross. 3.ii) Find the possible blood group of children if the blood group of father is O and that the mother AB.	2018
2.vi) Define Allele.	
2.x) What will be the result of a cross between Haemophilia- male and Normal-female 2.xii) Give definition of Gene pool OR Crossing over. 2.xiv) State the law of independent Assortment. 3. vi) Give the Cross between Red – eyed female and white – eyed male in drosophila.	2017
2. ix) Draw a chart showing Phenotype and Genotype of different Blood groups. 2. xii) A color blind women is married with a normal man. How many children of this couple are likely to be color blind in F ₂ generation? 2. xv) State the law of Segregation with Phenotypic ratio and Genotypic ratio.	2016

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2. v) Describe Test cross and Incomplete dominance.	2015
2. vii) Why did Mendal choose pea plant as an experimental material.	
2. iii) A color blind women marries a normal man. How many children of this couple are likely to be color blind in F ₂ generation	2014
2.vii) Draw a chart showing seven pairs of contrasting characters observed by Mendel. Also mention which of these characters are dominant or recessive.	
2. ix) What is a test cross? Explain with example.	
Section C	
4. Describe inheritance of two traits (Dihybrid Cross) with the help of checker board.	2025
4.—OR—Describe multiple allele with the help of ABO blood group.	
4. (OR) State the law of independent assortment. Explain with the help of a checker board.	2023
4. a) Explain multiple allele with the help of Human Blood Group.	2022
5. a) Describe sex linked inheritance in man with reference to color blindness.	
4. State Mendel's law of independent assortment with the help of checker board.	2021
5. Define Sex-linked inheritance. Explain with reference to haemophilia in man	2019
[6 OR] Describe the Law of Independent Assortment with the help of checker board.	2018
5. Describe the 'Law of Independent Assortment'. Explain it with the help of checker board method.,	2016
[OR 5] Define Sex-linked inheritance. Discuss Sex-linked inheritance in humans.	2015
3. State the Law of Independent Assortment. Explain it with the help of checker board method.	2014

Chapter: 24
EVOLUTION
Section B

iv) Define Homologous organs with example.	
v) Describe the Invagination theory.	2025
xi) Differentiate between Divergent and Convergent evolution.	
iii) Define Speciation. Name the types of Speciation.	2024

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2(n) vi) Why is Archaeopteryx considered a link between Reptiles and Birds?	2023
2. (a) (vii) How does artificial selection play role in the formation of new specie?	2022
2. b) (vii) Write any four objections on Darwin's theory of Natural selection.	2021
2.a) (iv) Why is Archaeopteryx considered a link between Reptiles and Birds?	2019
2.a) (iii) Why is Archaeopteryx considered a connecting link between Reptiles and Birds?	2018
2.b) (xv) State the Objections on Lamarck's Theory.	
2.b) (xiv) State the evidences of vestigial for evolution in man.	2017
(vi) What are Vestigial organs? Give the example with the reference of animals.	2016
(vi) Explain Hardy Weinberg Law.	2015
(iii) Write any four objections on Darwin's Natural Selection Theory.	2014
(xi) Give any two evidences in favour of Organic evolution.	
Section C	
3. Define Evolution. Discuss Lamarck's theory and the objection raised against it.	2016
5. Describe Darwin theory of Natural selection and the objections raised against it.	2015

Chapter: 25
MAN AND HIS ENVIRONMENT
Section B

vi) What are Aquifers? What is their role in the Earth's hydrological cycle?	2025
xii) Differentiate between Xerarch and Hydrarch succession.	
xiv) Write a short note on water cycle.	
ii) Describe the causes of ozone layer depletion OR Describe the causes of acid rain.	2024
ix) What are the advantages of nuclear power?	
ix) OR—Name two nuclear power plants of Pakistan.	
xiv) Describe the renewable and non-renewable environmental resources.	
2(a) vii) Why is algal bloom harmful for aquatic animals?	2023

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2. (a) (i) Why is depletion of ozone layer harmful?	2022
2. (b) (xiii) Write note on 'Algal bloom'.	
2.a) (v) How is Ozone layer being depleted?	2019
2.b) (xvi) Write note on Acid Rain OR Depletion of Ozone layer	2018
2.b) (xvi) How is Ozone layer being destroyed?	2017
(ii) Write a note on Lithotrispy or Acid Rain. (iv) Define renewable and non-renewable resources and examples.	2015
Section C	
5. What is meant by Biogeochemical Cycle? Describe Nitrogen cycle.	2025
5. What is meant by Biogeochemical Cycle? Describe Nitrogen cycle.	2024

Chapter: 26	BIOTECHNOLOGY
Section B	
vii) Why restriction enzymes are called molecular scissors? xiii) Define any two of the following: *Primary cell culture *3D culture *Cell line culture *Anther culture	2025
i) RFLPs and GMO are the abbreviations of terms iv) Define any two of the following terms: * Genome * Gel Electrophoresis * Tissue Culture * Genomic Library v) Describe the goal of Human Genome Project (HGP). vi) What is gene therapy? How many methods are there for gene therapy? Write their names. vii) Why restriction enzymes are called 'Molecular Scissors'?	2024
2(b) xiii) Define DNA fingerprint and how is it performed?	2023
2.a) (vi) Why is DNA fingerprinting performed? 2.b) (xv) State the significance of Genetically engineered bacteria.	2019
2.a) (vi) Why are plasmids used in Recombinant DNA. Technology?	2018
2.a) (vii) Why is DNA Finger Printing performed?	
2.b) (xiii) Write note on Transgenic animals.	2017
(xv) Write a note on Human Genome Project.	2016

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(v) Write the basis steps of recombinant DNA technology.	2015
(x) Write a note on DNA fingerprinting.	
(xv) Write a note on DNA fingerprinting.	2014
Section C	
6. Explain techniques of Gene Cloning by Recombinant DNA technology.	2025
3. What is PCR? Describe gene amplification through PCR. Writes its two applications.	2024
4. Explain various steps involved in recombinant DNA technology. Draw labelled diagram of different steps of insulin formation by this technology.	2023
5. Explain the various steps involved in Recombinant DNA technology and draw neat and labelled diagram	2019
[3 OR] Differentiate between Unifactorial and Multifactorial defects.	2018
5. Describe the different step in DNA recombinant Technology. Draw labelled diagrams of different steps.	2017
[3. OR] Write the various steps involved in recombinant DNA technology with the help of diagram.	2016
5. Write is Gene therapy?	2014

Chapter: 27	BIOLOGY AND HUMAN WELFARE
Section B	
viii) What is meant by home gardening? Write its significance. viii)—OR—Write the names of five vegetables and five fruits of summer season.	2024
Section C	
3.—OR— Write the role of microbes in Food Processing and Sewage Treatment.	2024
5.—OR—What is vaccination? Explain its role in preventing 'Hepatitis' and 'Tetanus'.	

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QUESTIONS	YEARS
Chapter: 15 HOMEOSTASIS	
Section B	
2(a) iv) Why star fish and jelly fish do not need Osmoregulation? 2(a) vii) How do Osmoregulation take place in terrestrial animals?	2025
2. (a)(i) How do aquatic osmoregulator animals overcome the osmoregulatory problems? 2. (b)(xi) Write the names, causes and symptoms of disorders of Urinary Tract.	2024
2(a) iv) How does osmoregulation take place in fresh water fish? 2(b) xi) Draw the Ornithine Cycle.	2023
2. (a) (iii) How are waste products removed from the body in case of renal failure? 2. (b) (xi) How do osmoregulation take place in terrestrial animals? 2. ix) Name various aspects of homeostasis	2022
2. i) Define Homeostasis. 2. vi) Why hydrophytes do not have cuticle on stem and leaves? 2. b)(x) Write the process of osmoregulation in terrestrial animal.	2021
2.a) (i) How does Osmoregulation take place in Trout fish? 2. (i) Name any two aspects of Homeostasis. 2.b) (xii) Write a note on Lithotripsy.	2019
2.a) (ix) How does Osmoregulation take place in Marine Bony Fish?	2018

2.b) (vi) Write note on two different types of Nephrons related to adaptation in kidney.	
2.a) (ii) How does fresh water fish regulate Osmoregulation?	2017
2. (iii) Name various aspects of homeostasis	
(ii) Explain osmoregulation in fresh water fish.	2016
(xiv) Define Dialysis. Write any one methods of its.	
(iii) Describe osmoregulation in Terrestrial animals.	2015
(xiv) Explain the role of Brain in the regulation of the body temperature.	

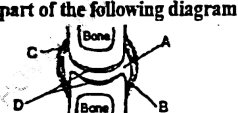
Section C

3. Define Thermoregulation. Explain its mechanism in human during hot and cold environment.	2025
5. What is Nephron? Describe the functions of nephron with the help of diagram.	2024
3.(a) Describe regulatory functions of Kidneys.	2022
5. (a) State Physiological mechanisms of thermoregulation in cold temperature.	2021
5. (b) Write the role of Brain in thermoregulation.	
3. Explain the Regulatory functions of the human Kidney and draw a labelled diagram of nephron.	2019
3. Explain the regulatory functions of human kidney and draw a labelled diagram of nephron.	2017

Chapter: 16 SUPPORT AND MOVEMENT**Section B**

2(a) i) Why calcium is essential for cross bridge formation? 2(a) iii) Why cardiac muscles not fatigue? 2(b) ix) Differentiate bone and cartilage, 2(b) x) Write a note on Sprain OR Spondylosis. 2(b) xii) Define joints. Name various types of joints.	2025
2. (a)(vi) Why are skeletal muscles get fatigued while cardiac muscles do not? 2. (b)(xiii) Write any two skeletal disorder of man.	2024
2(a) iii) Why is muscle fatigue common in sportsmen? 2(b) ix) Differentiate between skeletal muscles and smooth muscles.	2023
2. (iv) Why does abnormal muscle contraction occur in sportsmen?	2022
2. b)(x) State the functions of skeleton, 2. b)(ii) Differentiate between skeletal muscles and cardiac	2021

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muscles	2019
2.a) (ii) Why do Skeletal muscles show fatigue after heavy exercise?	
2.b) (xi) Write a note on Smooth muscles or Cardiac muscles.	
2.b) (xvi) What are the three main causes of Skeletal deformities? Give one example of each with symptoms.	
2.a) (iv) What is the cause of Muscle Fatigue?	2018
2.b) (xii) Write four differences between Bone and Cartilage.	
2.a) (iv) What is the reason of abnormal muscle contraction in sportsmen?	2017
2.b) (x) Differentiate between Skeletal muscles and Smooth muscles.	
(v) Write a note on Antagonistic muscles.	2016
(ix) Write a note on Abnormal muscular contraction.	
(vii) Write four functions of skeleton.	2015
(xi) Name the mark part of the following diagram.	
	
(xv) What is the function of Pelvic girdle? Name its three important bones.	
Section C	
3. —OR— Explain sliding filament model of muscle contraction and also describe cross bridge cycle with the help of labelled diagram.	2025
3. Define Skeletal Muscles. Describe the mechanisms of contraction of skeletal muscles with the help of diagram.	2024
5. Explain the Locomotion in any three of the following animals. Draw neat and labelled diagrams.	2023
a) Jelly fish b) Earthworm c) Snail d) Star fish	
4. Describe the mechanism of contraction in Skeletal muscle fiber.	2019
5. Describe Locomotion in Jellyfish, starfish and Earthworm.	2018
[3 OR] Define Skeletal muscles and describe the structure of Skeletal muscle fibre. Draw neat and labelled diagram of Skeletal fiber.	2017
4. Describe the structure of Striated muscles and explain their working mechanism with diagrams.	2016

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Chapter: 17	NERVOUS COORDINATION
Section B	
2(a) v) Why degenerative disorder of CNS is not curable?	2025
2. (a)(ii) Why receptors are called Transducers?	2024
2. (b)(xii) Write a note on autonomic nervous system.	
2. (a) (viii) How does Parkinson's disease different from Huntington's disease?	2022
2. a) ii) How sympathetic nervous system prepare the body for Fight or Flight response?	2021
2. b) xi) Write note on Reflex action.	
2.a) (viii) Why is Nervous co-ordination called the quickest way of communication within the body?	2019
2.b) (xiii) Define Receptors. Classify them with examples on the basis of stimuli.	
2.a) (viii) How is Parkinson's disease different from Huntington's disease? Write its treatment.	2018
2.b) (x) Write Four difference between Nervous and Chemical co-ordination.	
2.a) (vii) Why is Synapse termed as motor end plate?	2017
2.b) (xi) In which disease of the nervous system do seizures recur? State the Patient's condition in it.	
(iii) Write a note on Parkinson's disease.	2016
(iv) Differentiate between diffused nervous system and centralized nervous system.	
(vii) Write a note on Reflex action.	
(xii) Differentiate between Parasympathetic and Sympathic Nervous system.	
(xvi) Define Receptors. Name different type of receptors.	
(2.1) Write about the different types of Neurons.	2015
(ix) Write four Differences between Nervous coordination and Chemical coordination.	
(xvi) Describe the functions of Encapsulated receptors of the Skin.	
Section C	
4. Explain the structure and function of different parts of Human Brain. (diagram is not required)	2025
4. Define Nerve impulse. Explain resting membrane potential, action potential, depolarization, repolarization and synapse.	2024
3 (OR) Describe the structure and functions of different parts of human brain with neat and labelled diagram.	2023

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5.(a) Write nerve impulse under following heading: with the help of diagrams: • RMP • Propagation of Nerve Impulse	2022
3. (a) Write the structure and function of Human Brain (diagram is not required).	2021
[3 OR] Describe the structure and functions of different parts of the human brain. (Diagram is not required.)	2019
4. Describe the structures and functions of different part of human brain. (Diagram of brain is not required).	2017
4. Describe impulse under the following headings: a) AMP b) RMP c) SYNAPSE	2015

Chapter: 18 **CHEMICAL COORDINATION**

Section B

2(a) (i) Which hormones are released from adrenal medulla? Why they called emergency hormones?	2025
2(a) (viii) Why hormones secreted from adrenal cortex are called "Corticosteroids"?	
2(b) (xiii) Write any four functions of Hypothalamus.	
2. (a)(vii) Why does the pancreas function as both kinds of endocrine and exocrine gland?	2024
2. (b)(xiv) Name any four releasing and inhibiting hormones of Hypothalamus.	
2(a) (v) How does blood sugar level is controlled by pancreas?	2023
2. (a) (ii) Why Hormones secreted from 'adrenal medulla' are called 'emergency hormones'?	2022
2. (b) (xii) Write two disorders over, production of Growth Hormones (STH).	
2.a) (ix) How does pancreas regulate the amount of blood sugar?	2019
2.a) (ii) Which hormone is released from Adrenal Medulla and Why is it called Emergency hormone?	2018
2.b) (x) Write Four difference between Nervous and Chemical co-ordination.	
(iv) Differentiate between diffused nervous system and centralized nervous system.	2016
(vi) Write four Differences between Nervous coordination and Chemical coordination.	2015

Section C

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5. Define pituitary glands. Describe their lobes and secreted hormones.	2025
3. (b) Write the abnormal functions of Pituitary gland.	2021
[3. OR] Define Endocrine glands. Describe Pituitary gland as Master gland.	2015

Chapter: 19 **ANIMAL BEHAVIOUR**

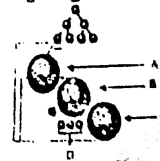
Section B

2. (a)(viii) Why three spined stickleback fish shows aggressive behavior?	2024
2. (b)(xv) What is taxis? Give their types with examples.	

Chapter: 20 **REPRODUCTION**

Section B

2(a) (vi) Why menstruation cease during pregnancy?	2025
2(b) (xi) Enlist the causes of infertility in male.	
2(b) (xiv) State the causes and symptoms of Gonorrhea.	
2(b) (xv) Describe miscarriage.	
2. (a)(iii) Why inner uterus wall thickens during secretory phase?	2024
2. (a)(iv) Why testes descend down in human male foetus before birth?	
2. (b)(ix) Differentiate miscarriage and abortion.	
2. (b) (ix) Write note on Gonorrhea or Syphilis.	2022
2. a) (v) Write two reasons which disturb menstrual cycle.	2021
2.b) (xvii) Draw labelled diagrams of different stages of Spermatogenesis or Oogenesis.	2019
2.a) (i) Why do Monozygotic twins have the same sex?	2018
2.b) (xiv) Write note o Gonorrhoea OR Syphilis.	
2.b) (xvii) Label the diagram given Below:	



2.1) Write a note on Spermatogenesis or Oogenesis.	2016
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(xiii) Write a note on Oestrous cycle.	2015
Section C	
4. ---OR--- Define menstruation. Describe its various phases and changes.	2025
3. Define Menstruation. Describe its various phases and its changes taking place in each phase with diagram.	2023
4. (a) Define Menstrual cycle and write its changes taking place in each phase.	2021
4. (b) Drawn labelled diagram of female reproductive organs	
[5 OR] Describe with the help of labelled diagram the human female reproductive system.	2019
3. Define Menstruation. Describe its various phases and the changes taking place in each phase with diagrams.	2018
[4 OR] Explain the process of Gametogenesis in human and draw a labelled diagram of male and female reproductive system.	2017
3. Define menstrual cycle. Describe its various phases and the changes occurring in each phase.	2015

Chapter: 21	DEVELOPMENT AND AGING	
Section B		
2. (a)(v) Why monozygotic twins having same sex?		2024
2. (b)(x) Write note on Rubella or abnormal Neural Tube Defects (NTDs),		
2(b) x) Write a note on Aging.		2023
2.a) (iii) How does Abnormal development take place in man?		2019
2.b) (xiii) Define Aging. Give its causes.		2018
(viii) Define Ageing and list its causes.		2016
Section C		